

ASESMEN_Pertemuan04

ASESMEN - PERTEMUAN 4 (S2)

JSON/CSV & Python

Informatika – Fase F / Kelas XI – SMA Negeri 6 Cimahi

Rubrik Penilaian

A. Struktur CSV & JSON (Bobot 15%)

Kriteria	1	2	3	4
CSV	Tidak benar	Format salah	Format benar	Benar + header
JSON	Tidak benar	Syntax error	Syntax benar	Benar + tipe data tepat

B. Baca CSV (Bobot 25%)

Kriteria	1	2	3	4
Kode berjalan	Error	Berhasil 1 fungsi	2-3 fungsi	4-5 fungsi
Hasil benar	0-1	2	3-4	5

C. DictReader & Filter (Bobot 20%)

Kriteria	1	2	3	4
Filter	Tidak jalan	1 filter	2 filter	3 filter + benar
Output	Error	Ada output	Output benar	Output + rapi

D. JSON (Bobot 20%)

Kriteria	1	2	3	4
Baca/tulis JSON	Tidak	Baca saja	Baca + tulis	Baca + tulis + modifikasi
Operasi data	Tidak	1 operasi	2 operasi	3+ operasi benar

E. Konversi CSV↔JSON (Bobot 10%)

Kriteria	1	2	3	4
Konversi	Tidak	1 arah error	1 arah benar	2 arah benar

F. Refleksi & Tugas (Bobot 10%)

Kriteria	1	2	3	4
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Kriteria	1	2	3	4
Refleksi	Tidak diisi	1 jawaban	2 jawaban	2 jawaban + mendalam
Tugas rumah	Tidak	Sebagian	Lengkap	Lengkap + Colab link

Kunci Jawaban

Soal 1 — CSV & JSON

CSV:

```
Nama,Kelas,Nilai UTS,Nilai UAS
Adi,X-A,82,88
Budi,X-B,75,70
Citra,X-A,90,92
```

JSON:

```
[
  {"nama": "Adi", "kelas": "X-A", "uts": 82, "uas": 88},
  {"nama": "Budi", "kelas": "X-B", "uts": 75, "uas": 70},
  {"nama": "Citra", "kelas": "X-A", "uts": 90, "uas": 92}
]
```

Soal 2 — Baca CSV (contoh)

```

import csv

with open('nilai.csv', 'r') as file:
    reader = csv.reader(file)
    next(reader) # skip header
    total_uts = 0
    total_uas = 0
    count = 0
    max_uts = 0
    min_uas = 100
    siswa_min_uas = ''

    for row in reader:
        nama, kelas, uts, uas = row
        uts = int(uts)
        uas = int(uas)
        total_uts += uts
        total_uas += uas
        count += 1
        if uts > max_uts:
            max_uts = uts
        if uas < min_uas:
            min_uas = uas
            siswa_min_uas = nama

    print('Jumlah siswa:', count)
    print('Rata-rata UTS:', total_uts / count)
    print('Rata-rata UAS:', total_uas / count)
    print('UTS tertinggi:', max_uts)
    print('UAS terendah:', min_uas, '-', siswa_min_uas)

```

Soal 3 — Filter DictReader

Filter	Jumlah	Nama
UTS $\geq$ 85	1	Citra
UAS < 75	1	Budi
Rata-rata $\geq$ 80	2	Adi, Citra

Soal 4 — JSON

```

# Tambah siswa
data.append({"nama": "Dian", "kelas": "X-C", "uts": 65, "uas": 70})

# Simpan lagi
with open('siswa.json', 'w') as f:
    json.dump(data, f, indent=2)

# Rata-rata UTS
total = sum(s['uts'] for s in data)
print('Rata-rata UTS:', total / len(data))

# Filter UAS > 80
lulus = [s for s in data if s['uas'] > 80]
print('Siswa UAS > 80:', [s['nama'] for s in lulus])

```

Soal 5 — Konversi (contoh)

```
# CSV → JSON
def csv_to_json(csv_file, json_file):
    import csv, json
    data = []
    with open(csv_file, 'r') as f:
        reader = csv.DictReader(f)
        for row in reader:
            data.append(row)
    with open(json_file, 'w') as f:
        json.dump(data, f, indent=2)

# JSON → CSV
def json_to_csv(json_file, csv_file):
    import csv, json
    with open(json_file, 'r') as f:
        data = json.load(f)
    with open(csv_file, 'w', newline='') as f:
        fieldnames = data[0].keys()
        writer = csv.DictWriter(f, fieldnames=fieldnames)
        writer.writeheader()
        writer.writerows(data)
```

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